

Matrix inequalities and norms

Ghabries's determinant comparison for arbitrary nonnegative base powers

Difficulty: challenging **Importance:** interesting to specialist

Status: Partially resolved

Rating rationale: The intermediate base powers resist the available determinantal comparisons, making the problem challenging; the immediate application is to specialist matrix-function inequalities.

Problem statement

For every integer $n \geq 1$, positive definite matrices $A, B \in \mathbb{C}^{n \times n}$, and real parameters $k \geq 0$, $0 \leq p \leq 2$, determine whether

$$\det(A^k + |AB|^p) \geq \det(A^k + A^p B^p)$$

holds. Here $|AB| = ((AB)^*(AB))^{1/2}$, and powers are defined by spectral functional calculus; the zeroth power of a positive definite matrix is the identity. Although the matrix inside the determinant on the right need not be Hermitian, its determinant is a positive real number: factor out A^k and use that a product of two positive definite matrices has positive eigenvalues.

The source states the conjecture for positive semidefinite inputs. The positive definite formulation captures the nontrivial question and avoids ambiguity in zeroth powers; positive-exponent semidefinite cases follow by regularization.

Such determinants arise in comparisons of interpolants for positive definite data. The distinction between a product's modulus and the product of matrix powers is important in matrix-function inequalities.

References

1. M. M. Ghabries, *Contributions to Matrix Inequalities and Some Applications*, PhD thesis, University of Angers and Lebanese University (2022), §2.5 and final Open Problems, Problem 1, pp. 109–110. [Thesis uploaded by its author.](#)
2. H. Abbas, M. M. Ghabries and B. Mourad, *New determinantal inequalities concerning Hermitian and positive semi-definite matrices*, *Operators and Matrices* 15 (2021), 105–116, §4, Conjecture 2, p. 116 (earlier, narrower parameter region). [Publisher PDF.](#)
3. M. M. Ghabries et al., *A proof of a conjectured determinantal inequality*, *Linear Algebra Appl.* 605 (2020), 21–28, main theorem. [Publisher.](#)

Status check (2026-09-10): the thesis retains the range $0 < k < 2$, $0 < p < 2$, after recording proved results for $k \geq 2$ and for $p = 2$. Its broader question supersedes the 2021 formulation. The 2020 resolution concerns Lin's earlier $k = 2$ question. The author's July 2026 normal-matrix determinant paper and searches for "Ghabries determinant arbitrary k conjecture", exact titles, and 2025/2026 yielded no full resolution. This is a bounded check.